

Demystifying Blockchain Technology and Cryptocurrency

DR. SAYED QAISER ALI SHAH
ASSISTANT PROFESSOR
FAST-NUCES, ISLAMABAD

Module 1

INTRODUCTION TO CRYPTOCURRENCY AND BITCOIN

History and Evolution of Money

- Visible money exists today is \$40 Trillion
- Invisible money is more than \$1300 Trillion
- 15000BC people lived in tribes
 - only aim pray and survive
- No pray daily, started Bartering system (Exchange of things)
- Bartering system started in Mesopotamiam tribes now Iraq
- People started trading

History of Cryptocurrency

- Problem raised with increase in population
 - Started writing on clay, Introduction of accounting
- Problem with barter system is that if you don't need one thing than why exchange?
- 2500BC commodity money was introduced like cow, alcohol, goat, swords, knife, tobacco and even human (Slaves)
- In yap Island people used heavy stones
 - weight was about 5 ton called Rai stones
- They remembered the ownership

History of Cryptocurrency

- Problem with commodity money is the worth of that?
 - can reduce the worth like alcohol expired
- 600BC metallic money in Ancient Greece (The Lydian Stater) → The first coin
- After Lydian Stater, every empire introduced their coin and started trading
 - difficult to carry large amount of coins
 - got stolen and counting problem

History of Cryptocurrency

- The era of paper money started in Mongolian china by emperor Kubla Khan, the grandson of Genghis Khan
 - Introduced paper money
 - Those who refused to trade with paper money were killed
 - This paper money was called IOU (I owe you)
- People thought why not use this for trading?
- Introduced banking system

History of Cryptocurrency

- People started exchange the gold with IOUs issued by people (Bank)
- IOU was issued in large amount and started inflation
- People contacted banks for their gold again not enough gold against that IOU
 - stopped trading against IOU
- 1816 Britain brought a law
 - Paper money against gold standard
- Raised an issue of deflation
- 1933 President of America Franklin D. Roosevelt ended the relationship between gold and money

History of Cryptocurrency

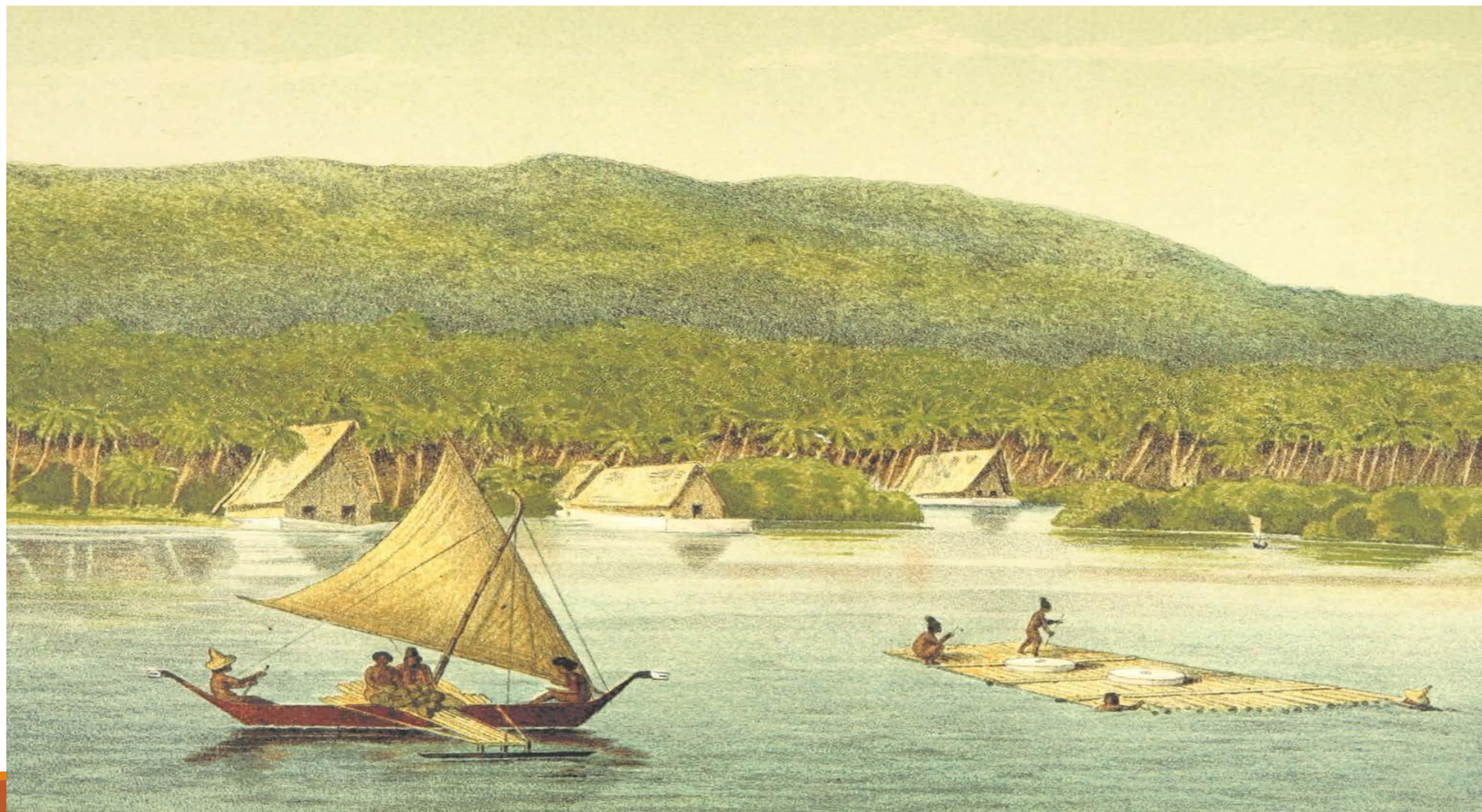
- Banks started publishing money like previous
- Relationship of money and gold is ended so money is just a piece of paper (FIAT money)
- 1980 E-Money was introduced to be transmitted immediately
 - The problem with digital money is hacking which is difficult but not impossible
- In 2009 Bitcoin was introduced which is cryptocurrency and is unable to get theft











Some Interesting Observations

- The stones themselves had no monetary value
- Spending your stone didn't require physically moving the stone - Just acknowledgement of a change of ownership
- Impossible to do a trade in secret
- They developed a form of distributed ledger but it couldn't scale

Cryptocurrency

- Definition: A digital or virtual currency secured by cryptography
- Key Features: Decentralized, Peer-to-peer, built on blockchain technology.
- The role of blockchain is securing transaction and providing transparency

Bitcoin (The first Cryptocurrency)

- Created in 2009 by **Satoshi Nakamoto**
- Motivation: Response to 2008 financial crisis
- Whitepaper: *“Bitcoin: A Peer-to-Peer Electronic Cash System”*
- No central bank, no trust needed → **Mathematical trust**

Bitcoin Underlying Protocol

- Peer-to-peer network
- Consensus mechanism: **Proof of Work (PoW)**
- Fixed supply: 21 million BTC
- New blocks every ~10 minutes
- Mining = solving complex puzzles

Activity/ Discussion



Prices Movements



Inflation



Deflation



Paper Currency vs
Cryptocurrency

Module 2

BLOCKCHAIN FUNDAMENTALS

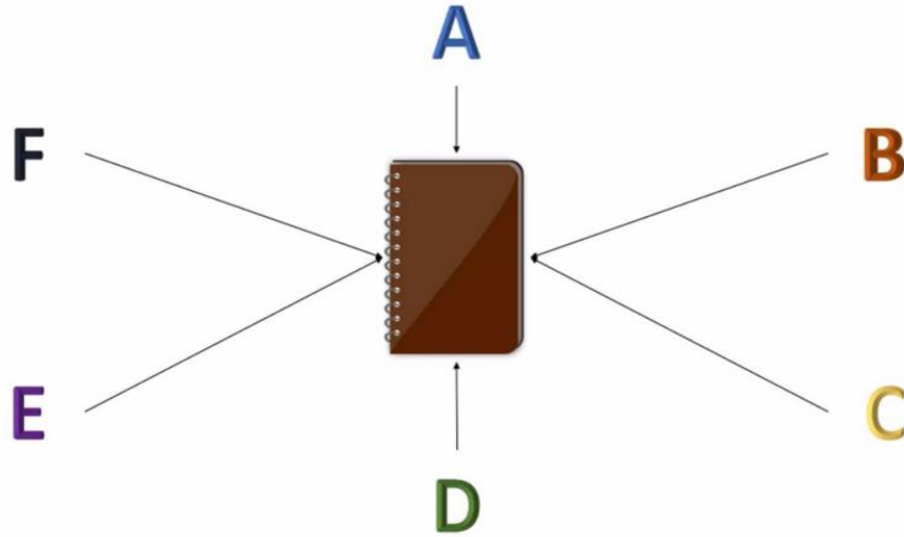
What is Blockchain

- Blockchain technology is a software; a protocol for the secure transfer of unique instances of values (e.g. Money, Property, Contracts and Identity Credentials) via the internet without requiring a third-party intermediary such as bank or government
- Email over IP, Voice over IP, Money over IP



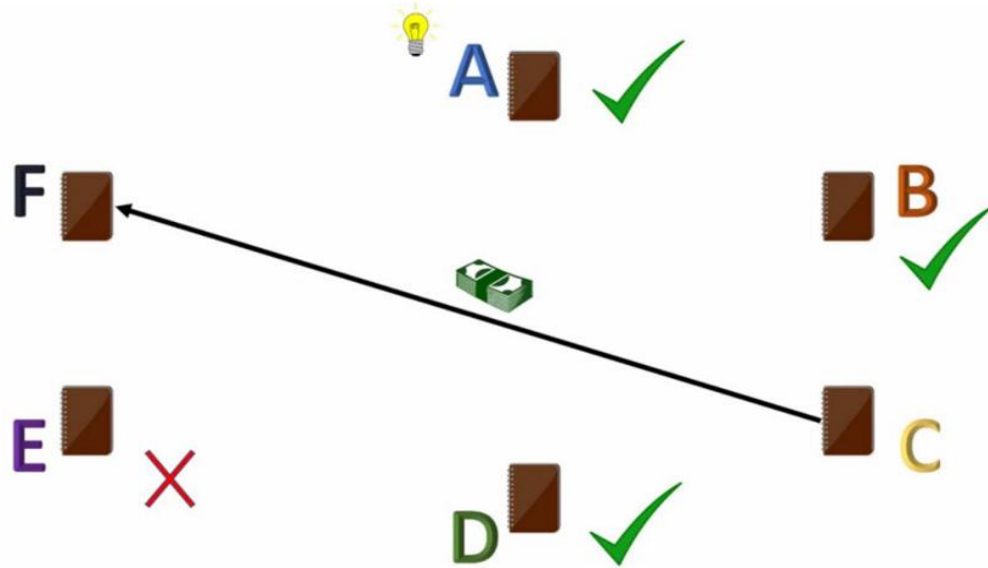
What is a Blockchain?

Common Ledger



What is a Blockchain?

⦿ A More Common Ledger



Evolution of Blockchain: Phase 1 - Transactions

- 2008-2013: Blockchain 1.0: Bitcoin Emergence
- Bitcoin vs Blockchain
- Blockchain Applications

Evolution of Blockchain: Phase 2 – Smart Contracts

- 2013-2015: Blockchain 2.0: Ethereum Development
- Vitalik Buterin
- Bitcoin vs Ethereum
- Smart Contracts

Definition?

Let's build our own definition

- It's a decentralized database which stores information in the form of transactions
- It can be public or private
- Stored data is Immutable
- Highly secure
- Data gets recorded via **consensus** based algorithms
- Uses cryptography
- Generally exist over peer-to-peer network

Definition?

So here is our definition:

“Blockchain is a consensus-based secure decentralized public/private database which stores information immutably over peer-to-peer network”

Definition?

- Each computer that participates in this P2P network is called a node. Each node maintains the records of transactions in multiple consecutive blocks.
- Blockchain technology is also called Decentralized Ledger Technology (DLT), as each node in the network keeps the same copy of the ledger. Please have a look at the following diagram



Chain of connected blocks

Need of Blockchain



Types of Blockchain

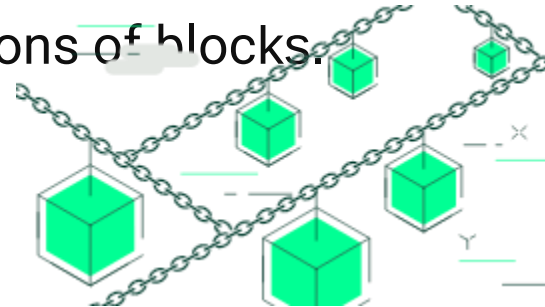
- Private vs Public
- Permissioned vs Permission less

Properties of Blockchain

- Distributed Ledger
- Fault Tolerance
- Attack Resistance
- Remove Intermediaries
- Consensus Protocol
- Low Transaction Fee
- Transparency
- Immutability
- Availability
- Timestamped

Blocks in Blockchain

- Each block on the chain comprises a number of transactions,
- and whenever a new transaction happens on the blockchain,
- a record of that transaction is added to the ledger of every
- participant. A blockchain network consists of millions of blocks.



Block Fields

1. **Header:** It is used to identify the particular block in the entire blockchain. It handles all blocks in the blockchain. A block header is hashed periodically by miners by changing the nonce value as part of normal mining activity, also Three sets of block metadata are contained in the block header.
2. **Previous Block Address/ Hash:** It is used to connect the $i+1^{\text{st}}$ block to the i^{th} block using the hash. In short, it is a reference to the hash of the previous (parent) block in the chain.

Block Fields

1. **Timestamp:** It is a system verify the data into the block and assigns a time or date of creation for digital documents. The timestamp is a string of characters that uniquely identifies the document or event and indicates when it was created.
2. **Nonce:** A nonce number which uses only once. It is a central part of the proof of work in the block. It is compared to the live target if it is smaller or equal to the current target. People who mine, test, and eliminate many Nonce per second until they find that Valuable Nonce is valid.

Block Fields

1. **Merkel Root:** It is a type of data structure frame of different blocks of data. A [Merkle Tree](#) stores all the transactions in a block by producing a digital fingerprint of the entire transaction. It allows the users to verify whether a transaction can be included in a block or not.

Types of Blocks

There are three type of blocks in blockchain

- Genesis Block
- Valid Block
- Orphan Block

Consensus Algorithm

PROOF OF WORK



The Nonce

Block No.- 6
Nonce:
Data: Kshitij->Rakesh 500 coins Raj->Bella 200 coins
Prev Hash: 0000AB23
Hash:

The Nonce

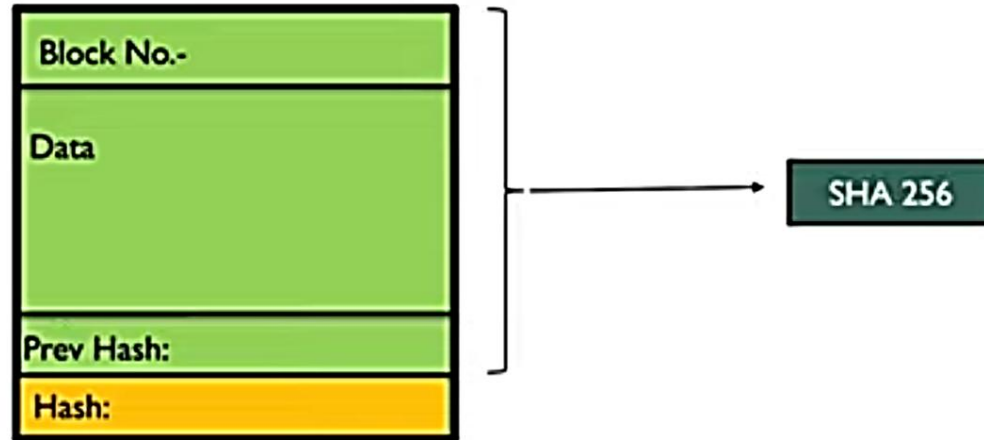
Block No.- 6
Nonce: 1001
Data: Kshitij->Rakesh 500 coins Raj->Bella 200 coins
Prev Hash: 0000AB23
Hash: 0000ef23

Hashing Algorithm



This has 64 hexadecimal characters.
Each character is of 4 bits.
So in total it has $64 * 4$ bits i.e. 256 bits.

Hashing Algorithm



BLOCKCHAIN WORKING



Genesis Block

Transactions In Blockchain

```
const transaction= {  
  nonce: " "  
  gasPrice: " "  
  gasLimit: " "  
  to: " "  
  value: " "  
  data: " "  
  v: " "  
  r: " "  
  s: " "  
}
```

Hash

0xfc42c91361f47191f9cdb
65e40f93be7fc44fedb9f27f
a48a8973c3287fd19e4

Merkle Trees

- Merkle tree also known as hash tree is a data structure used for data verification and synchronization.
- Merkle trees are useful in distributed systems where same data should exist in multiple places.
- A Merkle tree is a binary tree formed by hash pointers.

H1,H2,H3,H4 are hash value of T1,T2,T3,T4.

H1

H2

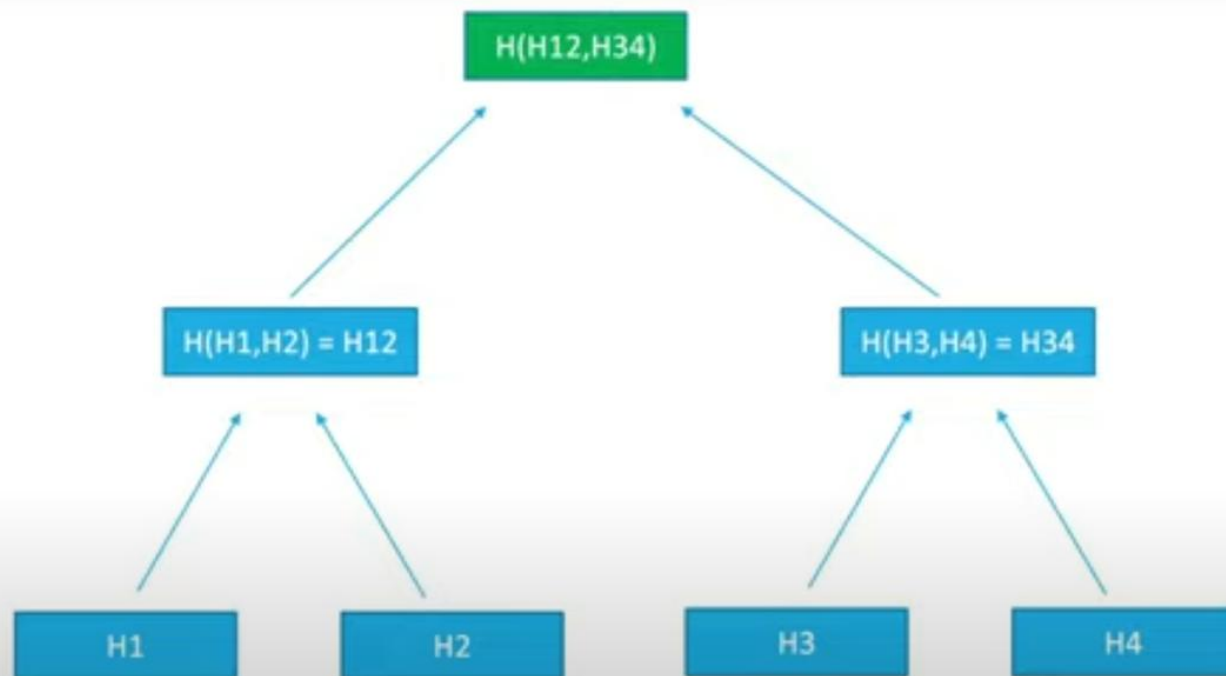
H3

H4

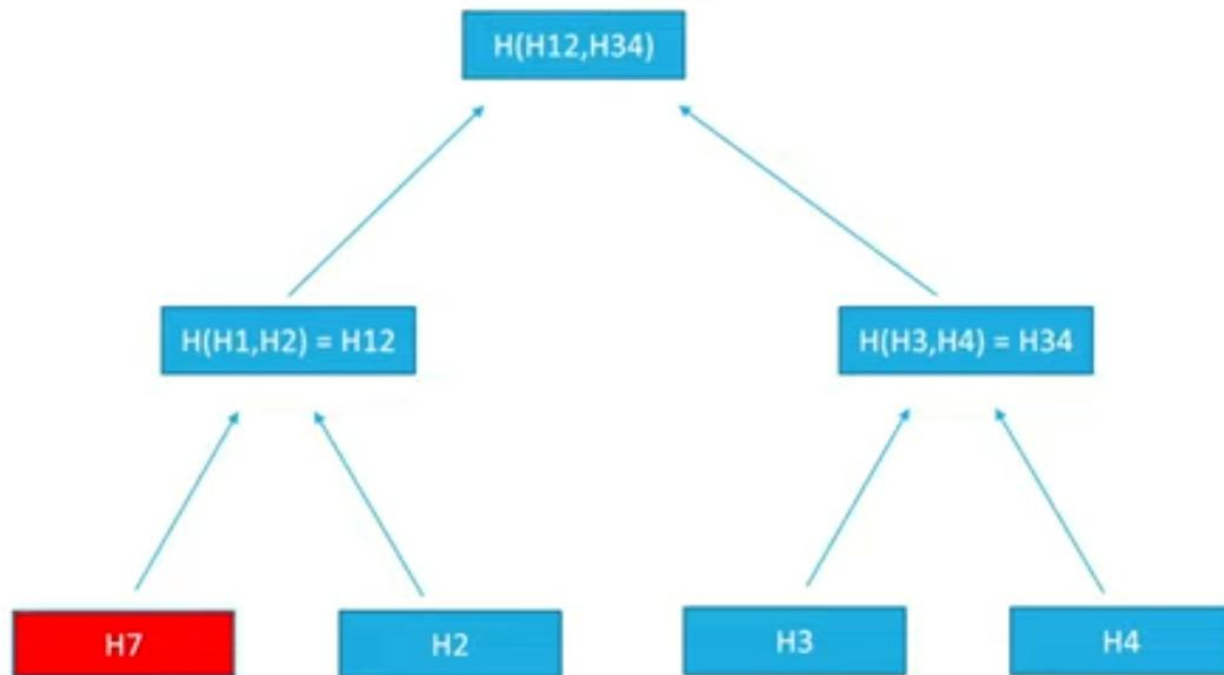
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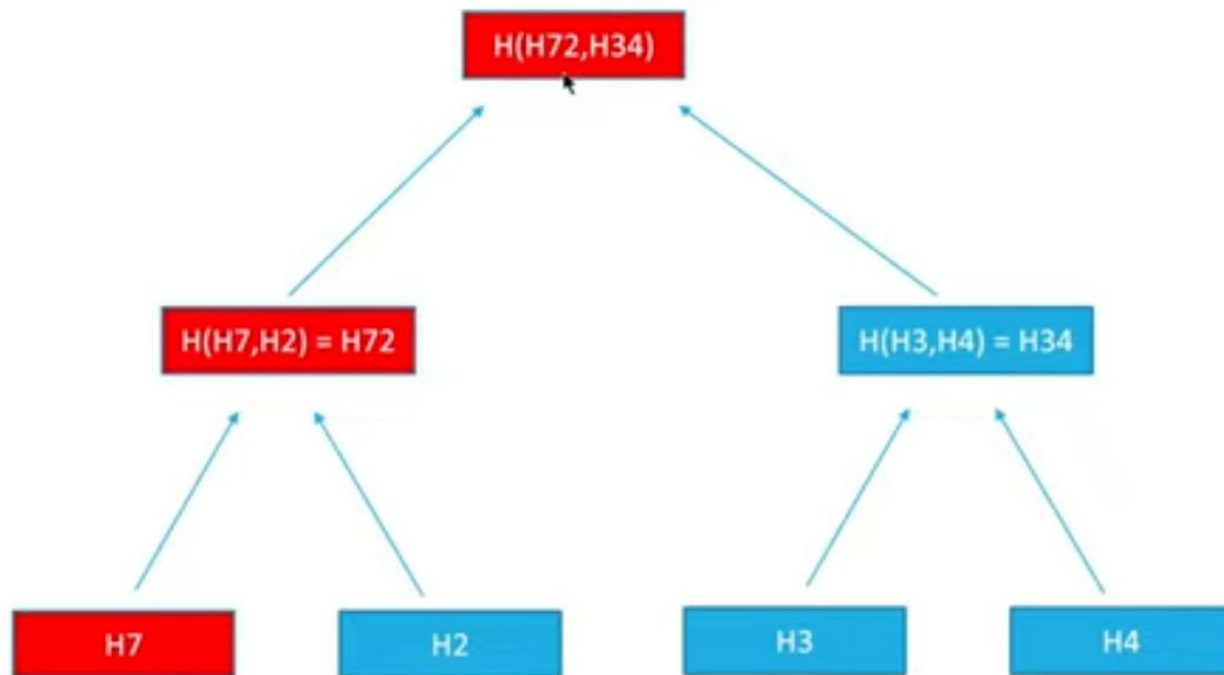
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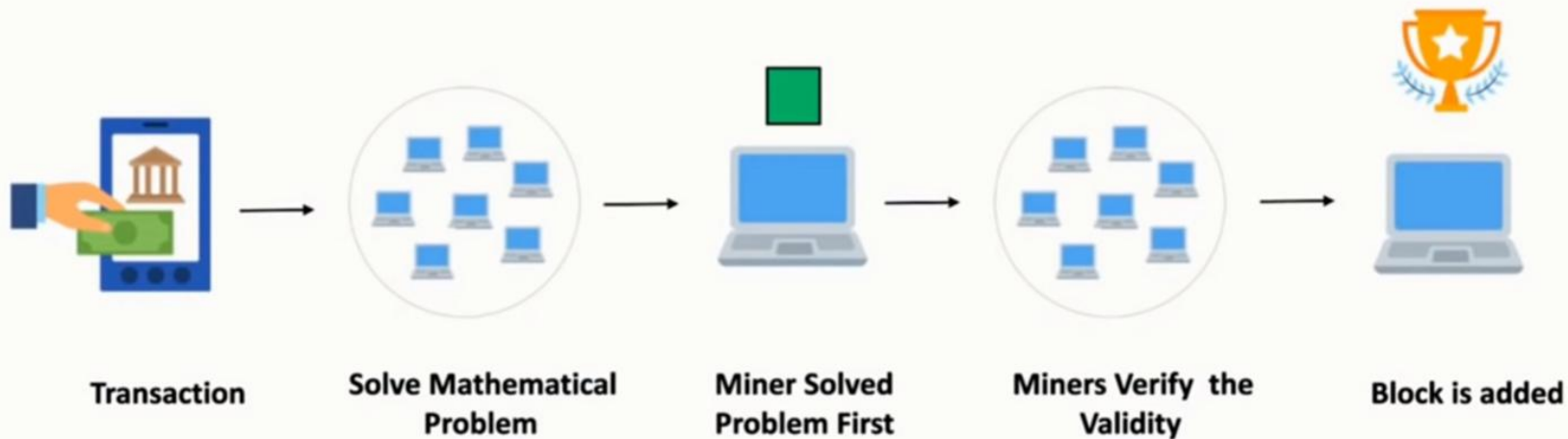
H1,H2,H3,H4 are hash value of T1,T2,T3,T4.



Advantages Of Merkle Tree

- 1) Efficient verification**
- 2) No delay**
- 3) Less disk space**

Blockchain Mining

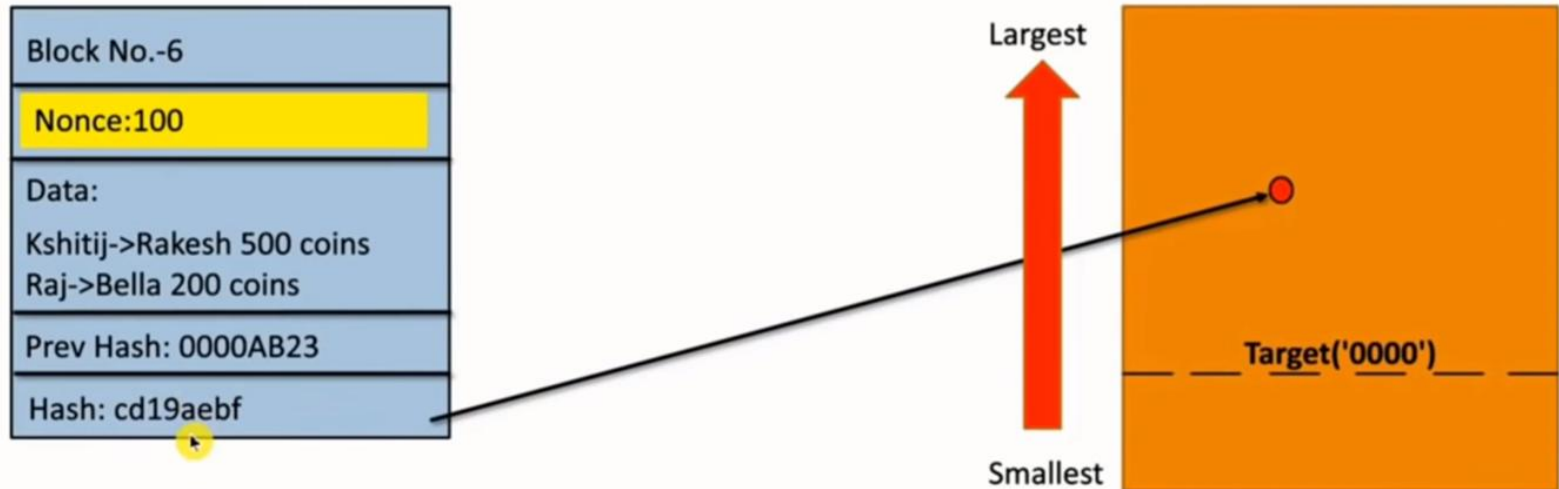


How Mining Works ?

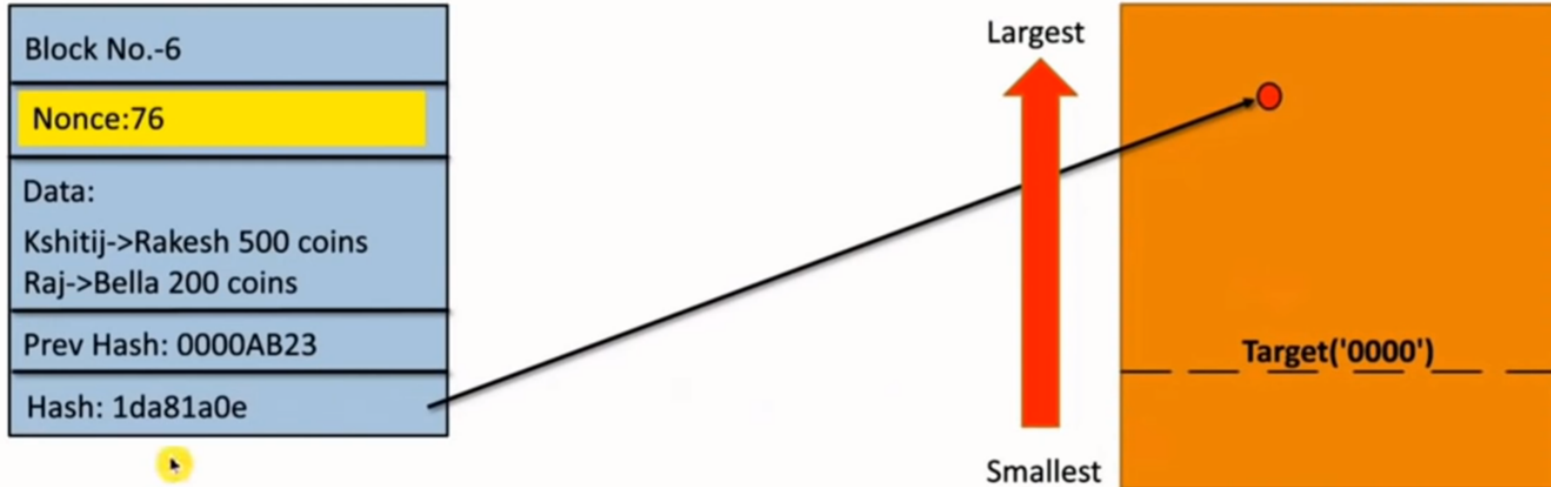
Target:

- Target is a number used in mining.
- It is a number that a block hash must be below for the block to be added on to the blockchain.
- The target adjusts every 2016 blocks (roughly two weeks) to try and ensure that blocks are mined **once every 10 minutes** on average.

How Mining Works ?



How Mining Works ?



How Mining Works ?

Block No.-6
Nonce: 201
Data: Kshitij->Rakesh 500 coins Raj->Bella 200 coins
Prev Hash: 0000AB23
Hash: b474e220

Largest

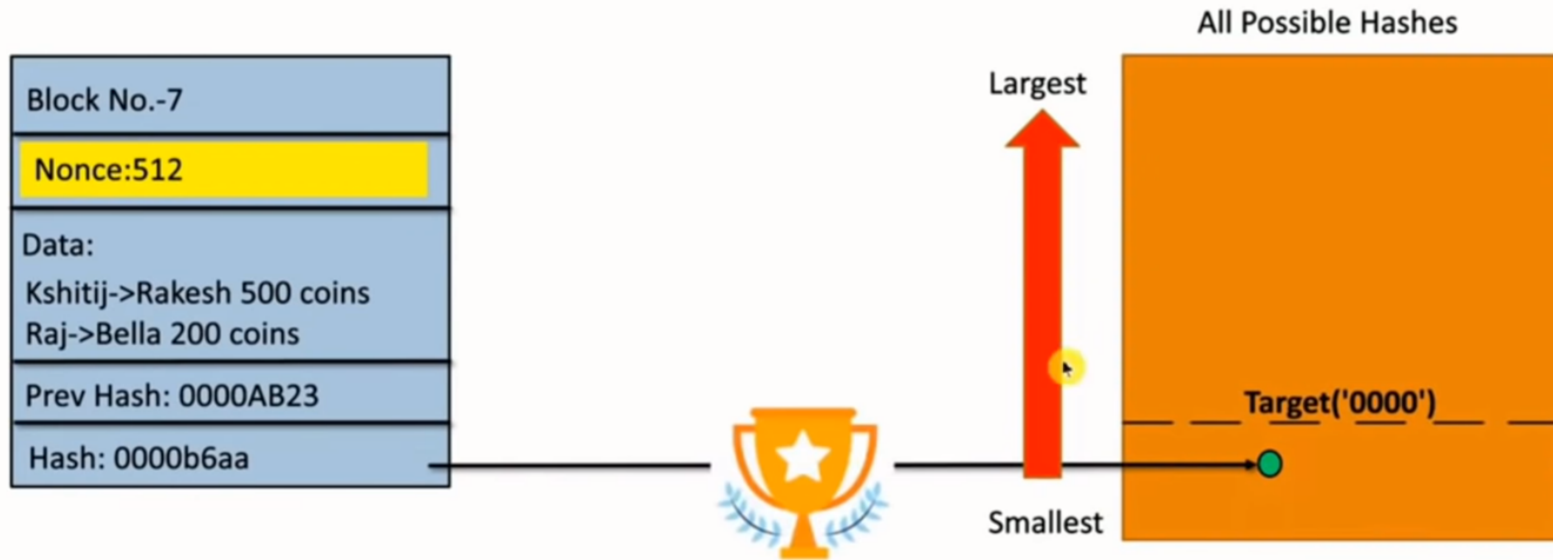
Smallest

All Possible Hashes

Target('0000')



How Mining Works ?



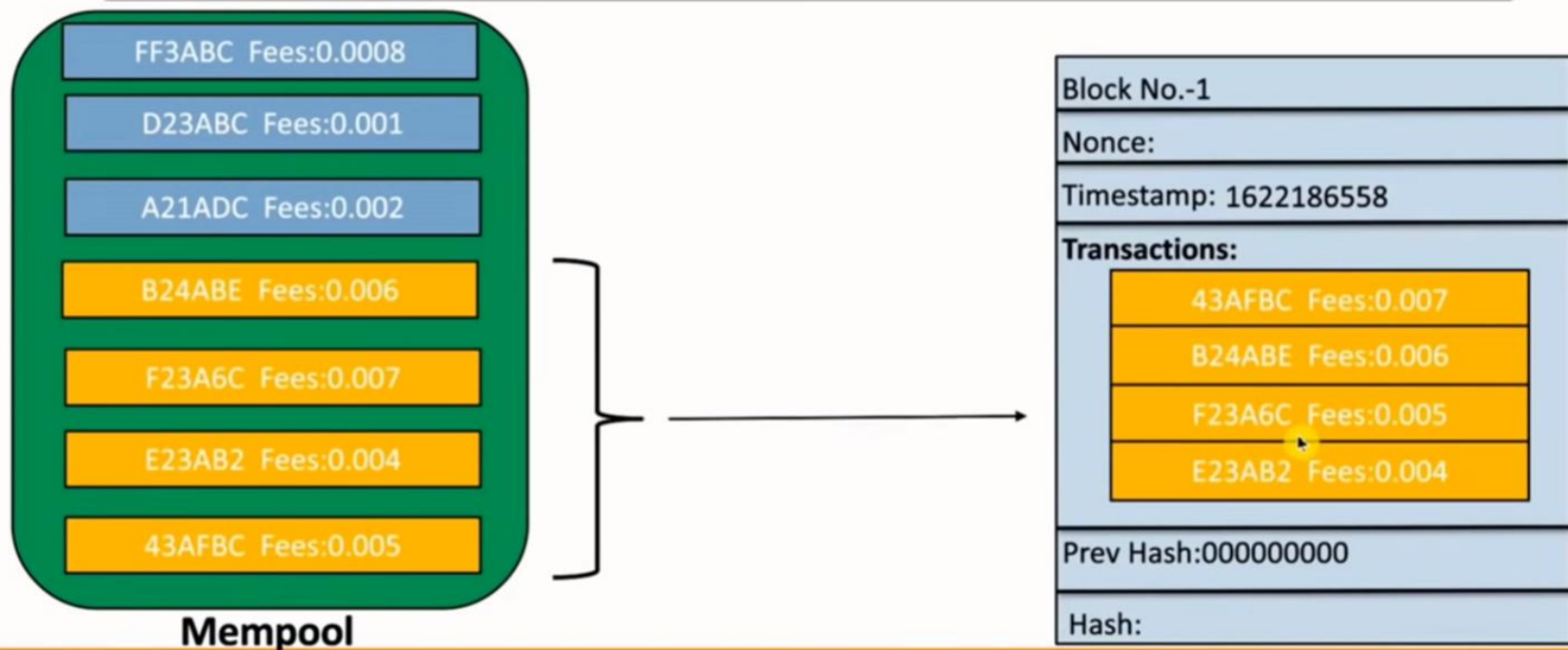
Mempool



Mempool

Block No.-1
Nonce:
Timestamp:
Transactions:
Prev Hash:000000000
Hash:

How actually mining of transaction takes place?



Bitcoin General Information

- Total Number of Blocks to be mined → 13.44 Million
- Estimated time for each block to mine → 10 minutes
- Block mined on 1st December 2024 at 3:00 PM → 872,605
- Last block will be mined in Year 2140
- Total BTC to be mined is 21 Million

A Historic Event – Bitcoin Pizza Day

- On **22nd May 2010** programmer named **Laszlo Hanyecz** made the first real-world transaction with Bitcoin by purchasing **two pizzas** from Papa John's.
- He paid **10,000 BTC** for the pizzas, which, at the time, was worth about **\$25**
- Price of 10,000 BTC today more than **Billion Dollars**
- **9th Feb 2011** Price of **BTC** was **\$1**

What we need for Mining?

ASIC Machines

Bitmain Antminer S21 XP (Air-cooled, 270 TH/s)

Hashrate: ~270 TH/s

Power Draw: ~3,645 W

Cooling: Standard air-cooled

Use Case: Suitable for mid-size or home miners with proper ventilation

Approx Price: Lower than hydro model (~\$4k-6k)

Revenue: \$15-\$16 per day before Electricity Cost

What we need for Mining?

Laptop

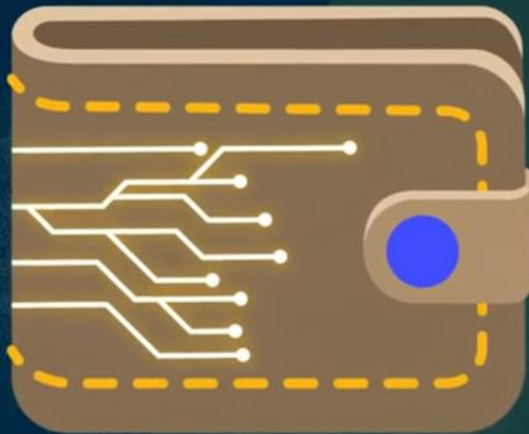
- Example: Tari Universe

Cell Phones

- Pi Mining

Blockchain Key-Pair

PUBLIC KEY



PRIVATE KEY

Cryptography



Sender



Hi Friend !!!



Receiver

Cryptography



Sender

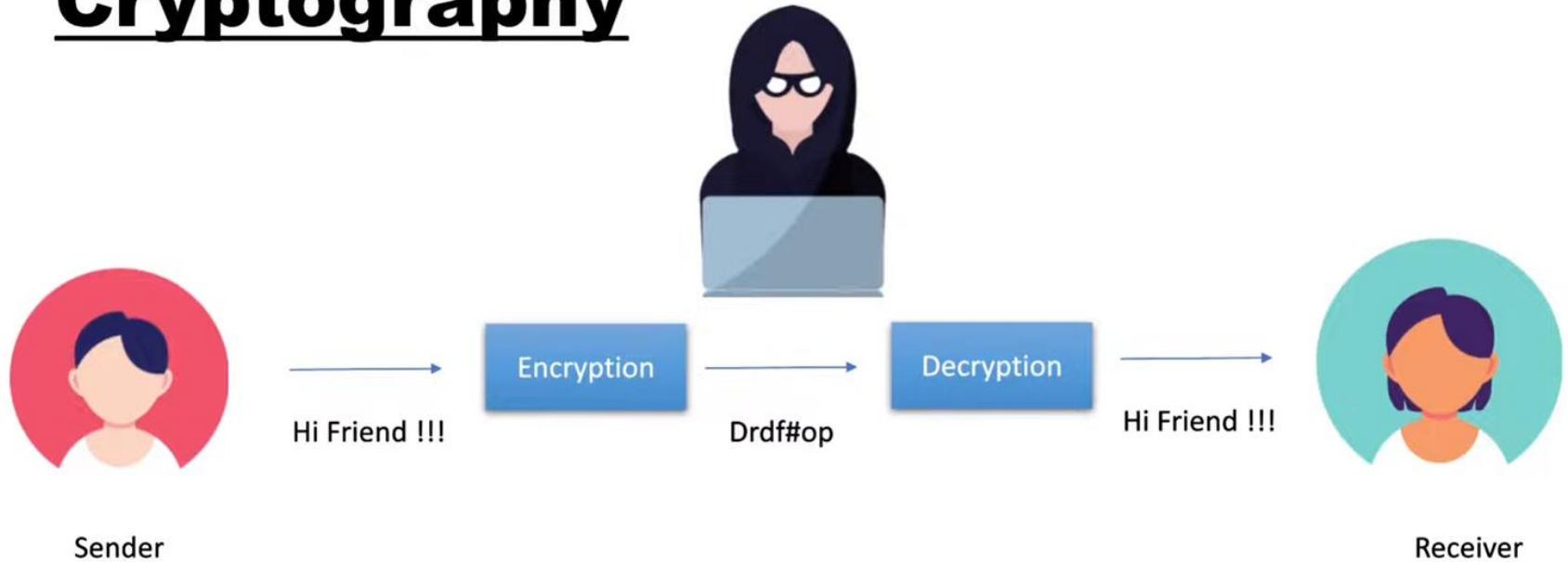


Hi Friend !!!



Receiver

Cryptography

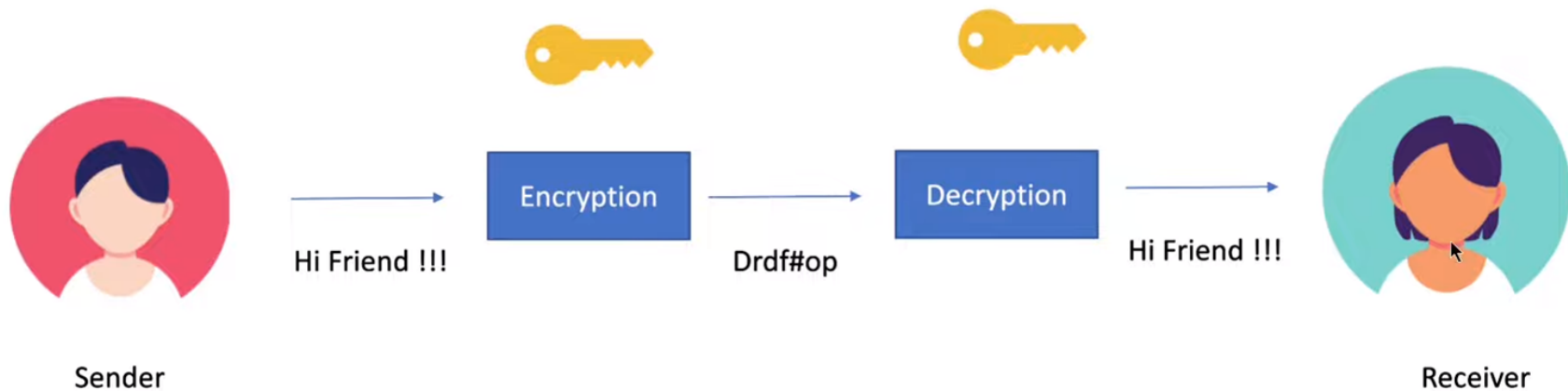


Two Types

Symmetric

Asymmetric

Symmetric Key Cryptography



Asymmetric Key Cryptography



Asymmetric Key Cryptography

A's Public Key



Asymmetric Key Cryptography

A's Private Key



Asymmetric Key Cryptography



A's Public Key



B's Public Key



A's Private Key



B's Private Key



Sender



Receiver

Asymmetric Key Cryptography



B's Public Key



A's Public Key



Private Key



Private Key

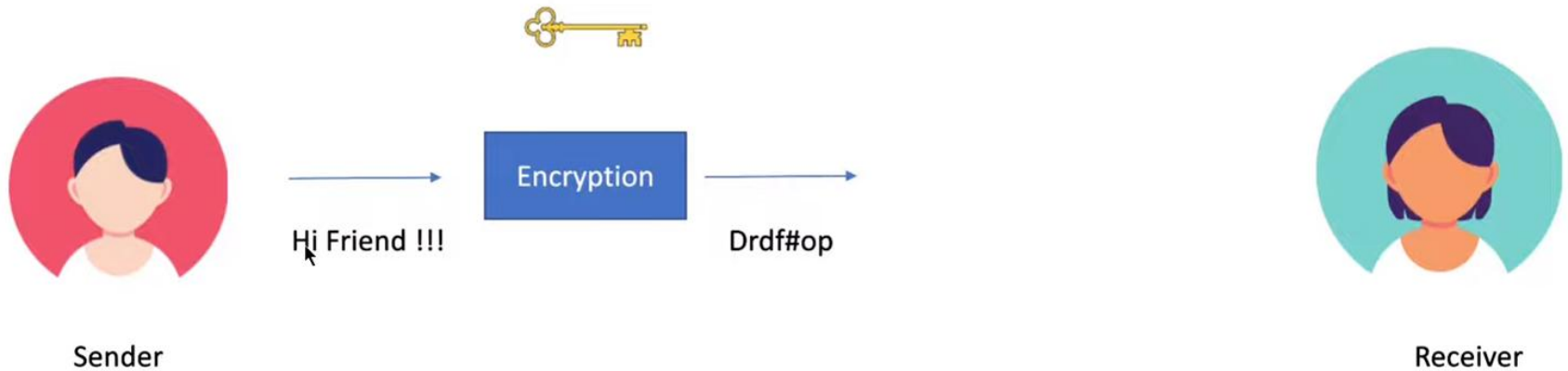


Sender

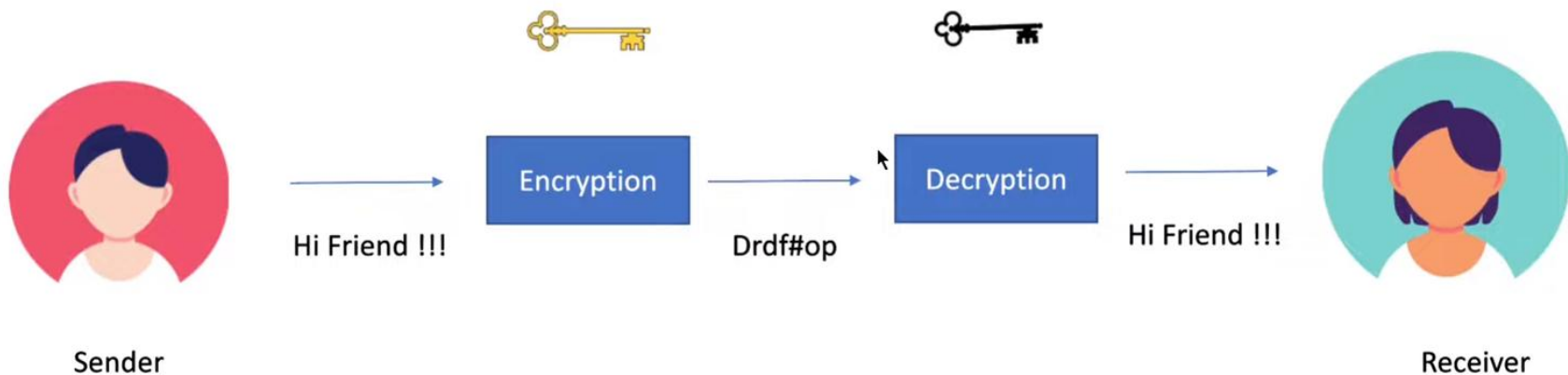


Receiver

Asymmetric Key Cryptography



Asymmetric Key Cryptography

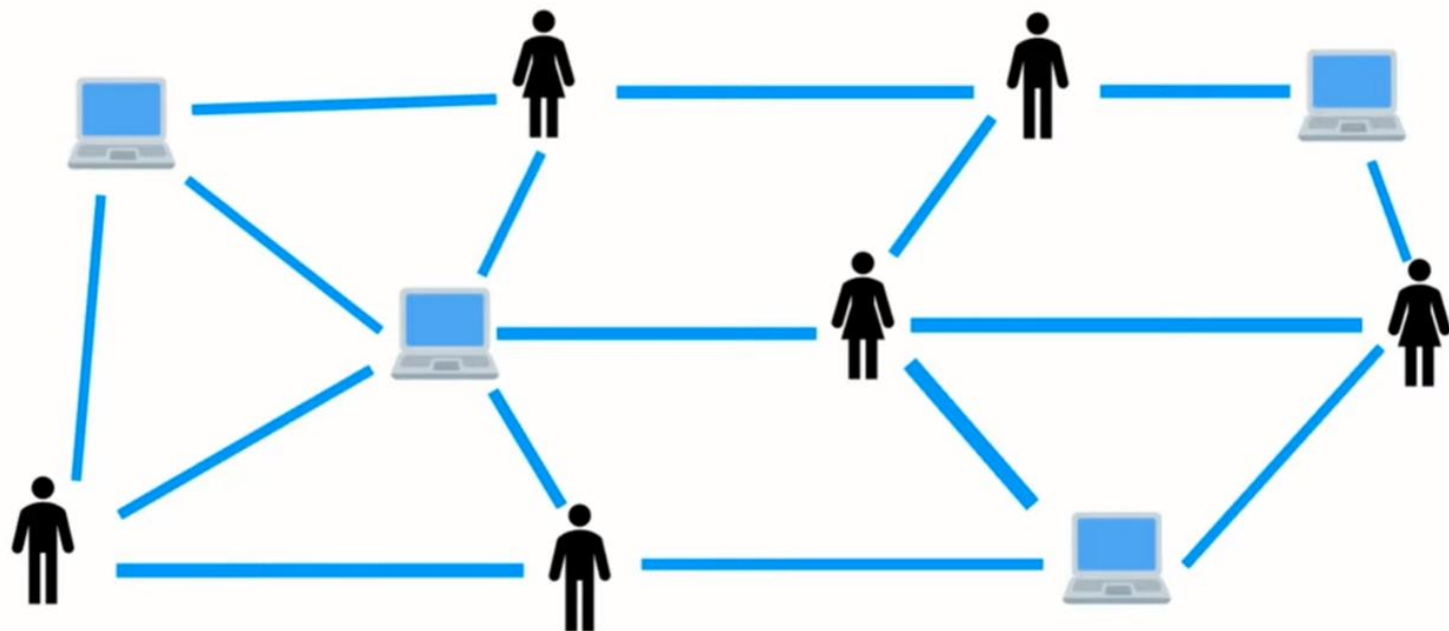
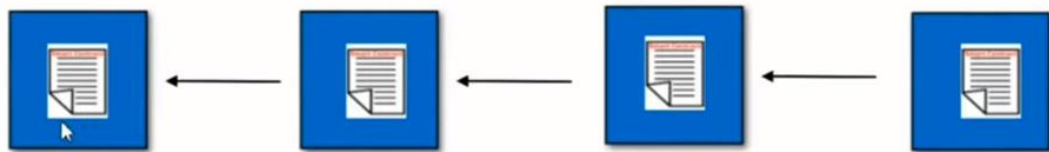


Ethereum



Vitalik Buterin





What is Ethereum?

- **Ethereum** is an open-source blockchain-based platform.

Bitcoin



Ethereum





Proof of Stake



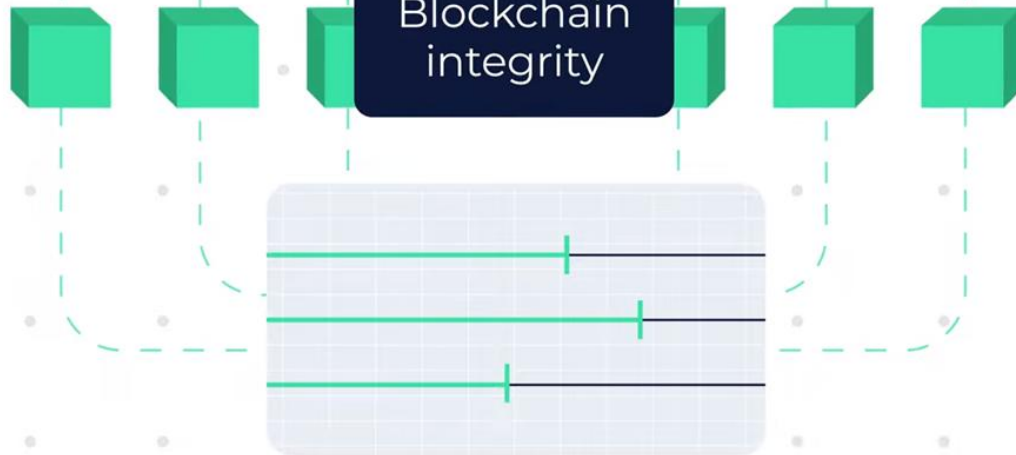
Proof of Work



Proof of Stake

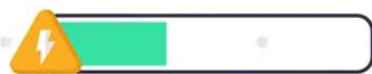


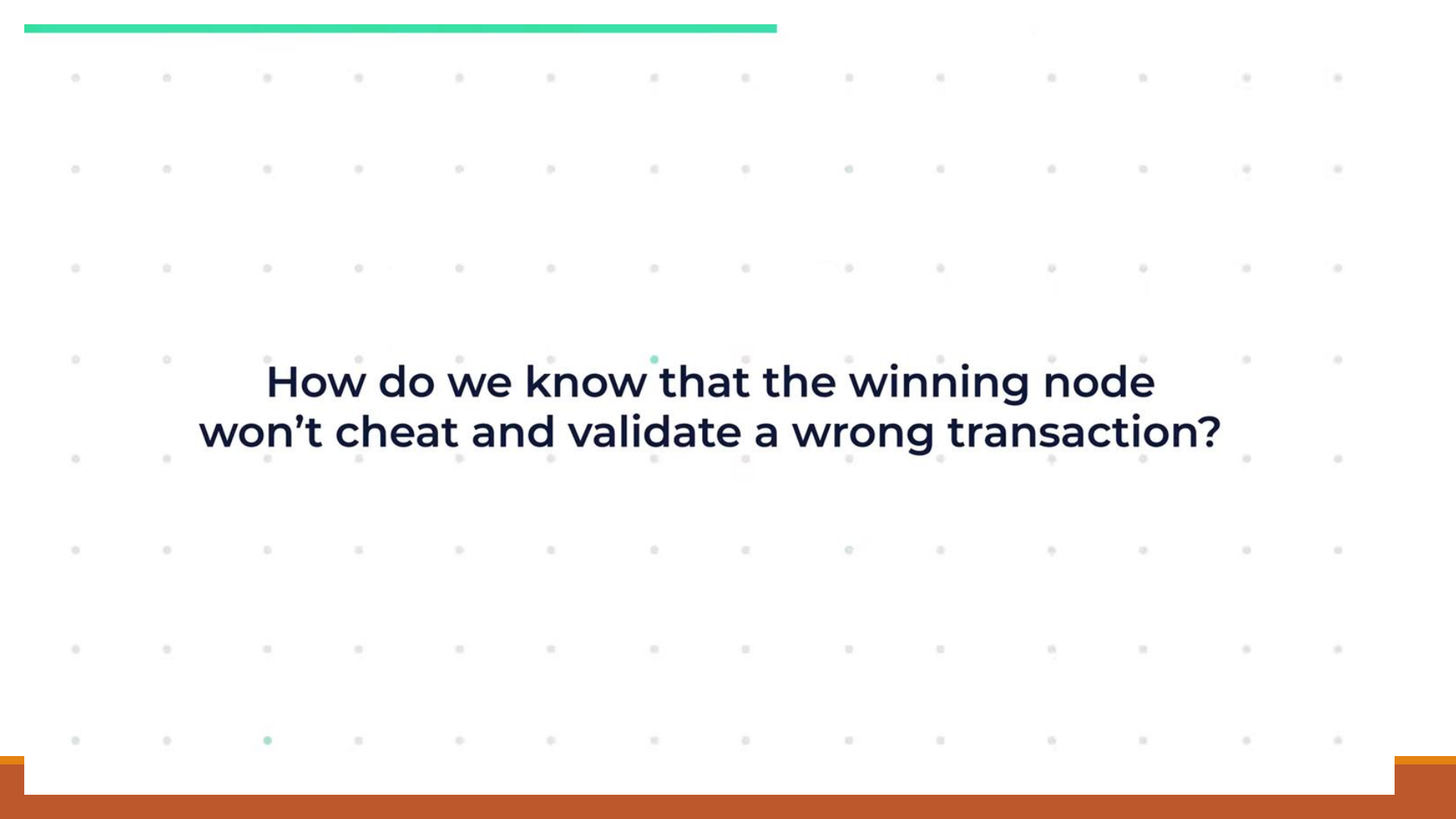
Blockchain
integrity









The background features a light gray grid of dots. A solid green horizontal bar is at the top left, and a solid orange horizontal bar is at the bottom right.

**How do we know that the winning node
won't cheat and validate a wrong transaction?**







Proof of Stake

Pseudo-random
election process



Staking age

Randomization

Node's wealth



Proof of Stake

Pseudo-random
election process



A decorative background featuring a grid of small dots. Most dots are light gray, but there are a few colored ones: a green dot at the top left, a teal dot in the upper middle, and a green dot at the bottom right. Additionally, there are solid colored bars: a green bar at the top left, a green bar under the word 'Proof' in the title, and orange bars at the bottom left and bottom right corners.

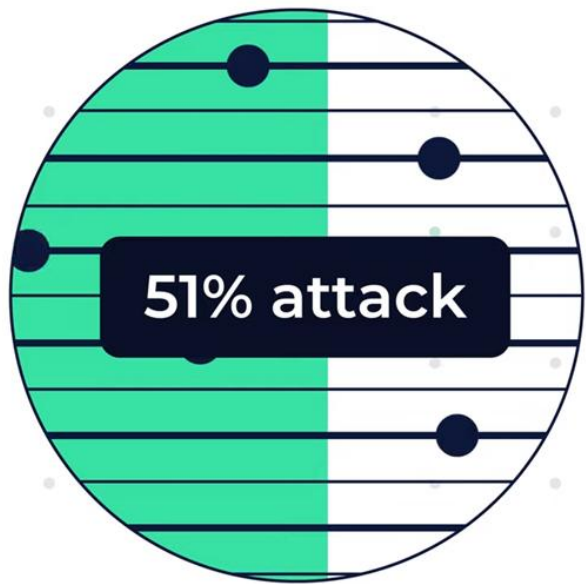
Security of Proof of Stake



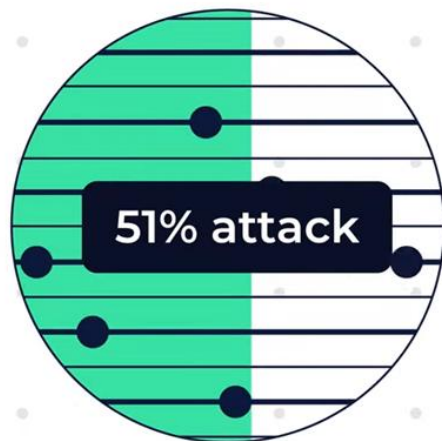
Avoid validating or initiating fraudulent transactions.







Required to successfully control the network and forcefully accept fraudulent transactions.



Proof of stake

Features of Proof of Stake

A decorative background featuring a grid of small dots. Most dots are light gray, but there are a few colored ones: a small green square in the top-left corner, a small green dot on the left side, a small green square below the title, and a small green dot in the bottom-right corner. The title text is centered in a dark blue font.

Advantages of Proof of Stake

The background features a light gray grid of dots. A solid green horizontal bar is located in the top-left corner. A solid orange horizontal bar is at the bottom, with a small orange square in the bottom-left corner.

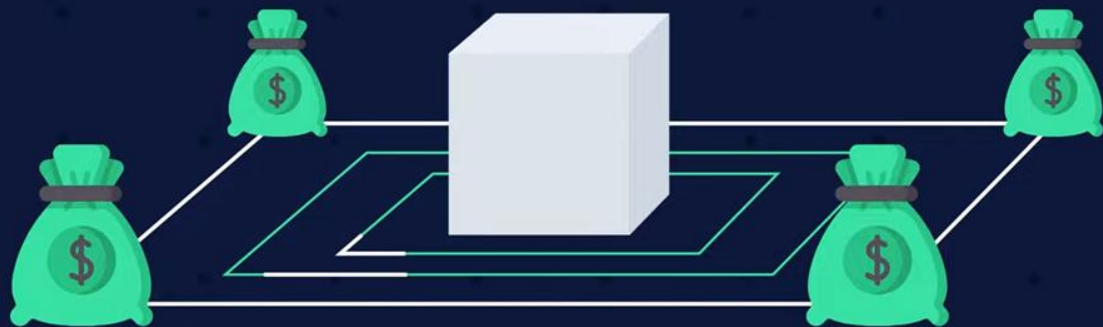
Energy-efficient

A grid of small dots arranged in 10 rows and 15 columns. Most dots are light gray. Two dots are highlighted in green: one at row 4, column 1 and one at row 7, column 7. The top-left corner has a solid green horizontal bar. The bottom-left and bottom-right corners have solid orange shapes.

Truly Decentralized

Disadvantages of Proof of Stake

**Validators with a Large Stake
can create an Oligopoly**



The network may grow more centralized over time.

Consensus Algorithms

- Proof of Work
- Proof of Stake
- Delegated Proof of Stake
- Proof of History
- Proof of Authority
- Proof of Burn
- Proof of Capacity
- Proof of Importance etc.

Coin vs Token

- Coin
 - A coin is a native cryptocurrency that operates on its own blockchain. It is a digital representation of value and serves as a medium of exchange within its respective blockchain network.
- Token
 - A token is a digital asset created on an existing blockchain. It does not have its own independent blockchain but exists on a platform that supports the creation of tokens through smart contracts.

Activity



Proof of work?



Proof of Stake?



Which one is better?



Applications of Blockchain?

Module 3

CRYPTOCURRENCY WALLETS AND EXCHANGES

Wallet



[Cryptography](#)

Choosing an Ethereum Wallet

- Wallet means a software application that manages your Ethereum Account.
- It holds your keys and can create and broadcast transactions on your behalf.
- If you choose a wallet and don't like you can change wallets quite easily.
- Make a transaction to send your funds from the old wallet to the new.
- Export your private keys and import them into the new one

Types of Wallet

Cold Wallet

Hot Wallet

- Mobile Wallet
- Desktop Wallet
- Web-based Wallet

Generally, the more popular a wallet application is , the more trustworthy it is likely to be

It is good practice to avoid “putting all your eggs in one basket” and have you Ethereum accounts spread across a couple of wallets.



Some good starter wallets

Metamask

- Browser extension wallet that runs in your browser
- It is easy to use and convenient for testing
- Able to connect to a variety of Ethereum nodes and test blockchains.



Some good starter wallets

Jaxx

- Multiplatform and multicurrency wallet that runs on a variety of Operating Systems
- It is often a good choice for new users as it is designed for simplicity and ease of use.
- Jaxx is either a mobile or a desktop wallet



Some good starter wallets

MyEtherWallet (MEW)

- Web-based wallet that runs in any browser.



Emerald Wallet

- It is an open-source desktop application
- Can run a full node/ connect to public remote node, working in light mode.
- It also has a comparison tool to do all the operations from the command line



Types of Exchanges

Centralized Exchanges

Decentralized Exchanges

Centralized Exchanges

- [Binance](#)
- [Bitget](#)
- [OKX](#)
- Bybit
- Coinbase etc

Decentralized Exchanges

Uniswap (Ethereum Blockchain)

PancakeSwap (Binance Smart Chain)

Trader Joe (Avalanche)

Raydium (Solana)

Activity

Transaction from Wallets

Use Metamask testnets and initiate Transactions

Creation of Binance Account

Overview of Binance Platform

Binance Web3 Wallet vs Binance Custodial Wallet

Adding Funds to Binance

Selling Funds through Binance

Module-4

MARKET STRUCTURE AND ORDER TYPES

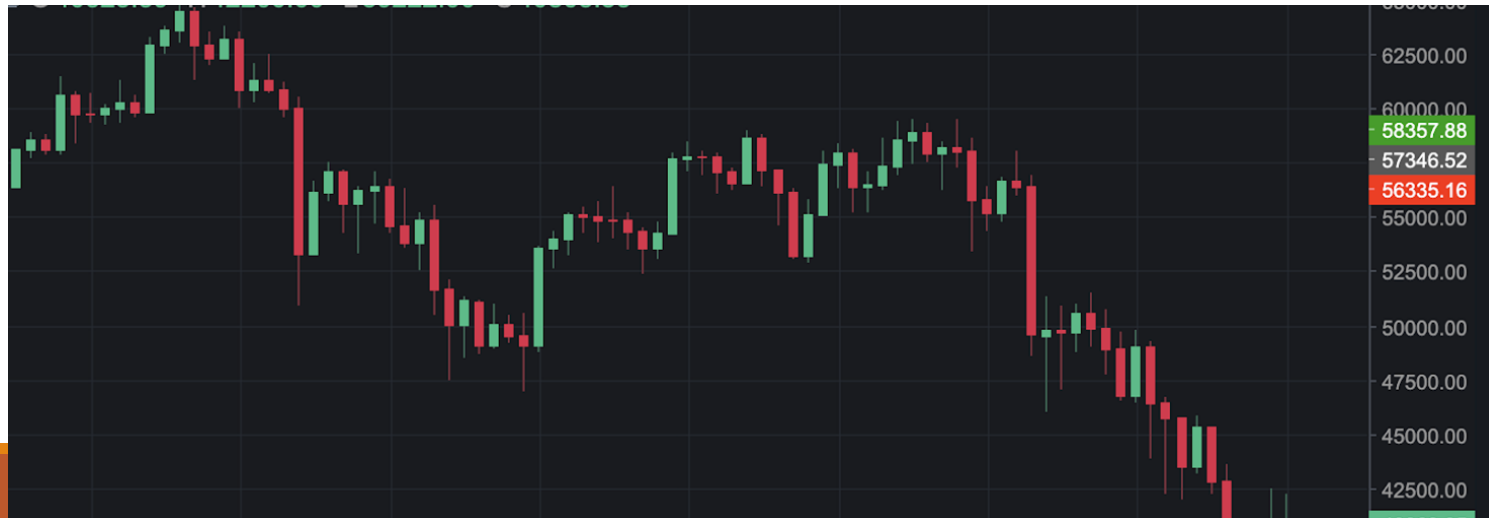
Trading

-
- **Buying and Selling of Cryptocurrency**
 - **Types of Trading**
 - Spot Trading
 - Futures Trading

Spot Trading

- **Buy** currency when the price is **Low**
- **Sell** it when the price is **high**

E.g. Gold



Why does the price go Up or Down?

SPOT TRADING DEMO

- Deposit and Withdraw Funds
- Buy and sell Cryptocurrency
- Trading Fee

Future Trading

- Traders can bet on the long and short positions with leverages
- Long → Bet that the price will go up (Pump)
- Short → Bet that the price will go down (Dump)
- Hedge → Opening both Long and short positions on a single coin at a time

FUTURE TRADING DEMO

- Taking Long and Short Positions
- Margin Modes
 - Isolated
 - Cross
- Leverage
- Liquidation
- Set Take Profit/ Stop Loss (TPSL)

Order Types

- Market Order
- Limit Order
- Take-Profit Order
- Stop-Loss Order
- Stop-Limit Order

Liquidity

How easily and quickly an asset (like BTC) can be bought or sold at stable prices without causing significant price movement.

Fast Transactions – Easy entry/exit.

Price Stability – Prevents large swings.

Lower Slippage – Minimal difference between expected and executed price.

Slippage

Difference between the **expected price** of a trade and the **actual executed price**

Why Does Slippage Happen?

Low Liquidity – Not enough buyers/sellers at desired price.

High Volatility – Prices change rapidly during execution.

Large Orders – Big trades move the market price.

Activity

Mock Trading Both Spot and Future
Risk Identification Exercises

Module-5

INTRODUCTION TO FUNDAMENTAL AND
TECHNICAL ANALYSIS

Fundamental Analysis

Evaluating a project's **true value** or **intrinsic worth** by studying **non-price data**

- Whitepapers
- Team and Advisors
- Tokenomics
- Use Case and Utility
- Roadmap and Development Activity
- Community and Social Media
- Partnerships and Ecosystem

Fundamental Analysis

Example: Bitcoin (BTC)

Use case: Digital gold, decentralized store of value

Team: Satoshi Nakamoto (anonymous)

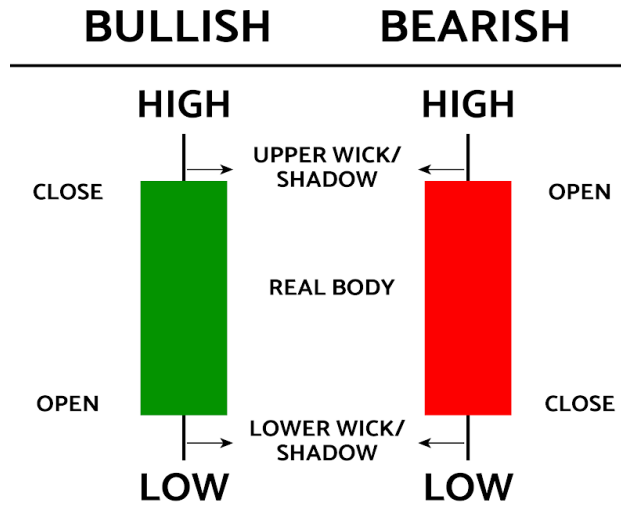
Tokenomics: Fixed supply (21M), halving every 4 years

Adoption: Most accepted crypto worldwide

Network: Highly secure, most miners and nodes

Development: Actively maintained on GitHub

CANDLESTICK BASICS



Reading Candlesticks Pattern



Trends

Bullish Trend

Bearish Trend

Sideways

Market Analysis

- Technical Analysis
- Sentiment Analysis

Technical Indicators

- **Price**
- **Volume**
- **Open Interest (OI)**
- **OI Notion Value (OI NV)**
- **RSI**
- **Support and Resistance**
- **Liquidation Chart**
- **Funding Rate**
- **MA**
- **Fear and Greed**
- **Long/short ratio**
- **Orderbook**

Activity

Pick a coin and perform Fundamental

Module-6

KEY INDICATORS AND CHART PATTERNS

Technical Indicators

- **Price**
Observe Opening and Closing Price of The candles
- **Volume**
The number of units traded during a time period
- **Open Interest (OI)**
Total Number of Open Contracts that are not settled
- **OI Notion Value (OI NV)**
How much actual money in USD is invested in Contracts

Cheat sheet

Price	Volume	OI	Interpretation
▲ Up	▲ Up	▲ Up	Strong bullish breakout
▲ Up	▲ Up	▼ Down	Short squeeze rally , might exhaust soon
▲ Up	▼ Down	▲ Up	Cautious bullish , weak buyers
▼ Down	▲ Up	▲ Up	Strong bearish breakdown
▼ Down	▲ Up	▼ Down	Long liquidation , panic selling
▼ Down	▼ Down	▼ Down	No conviction , ranging or ending trend

Technical Indicators

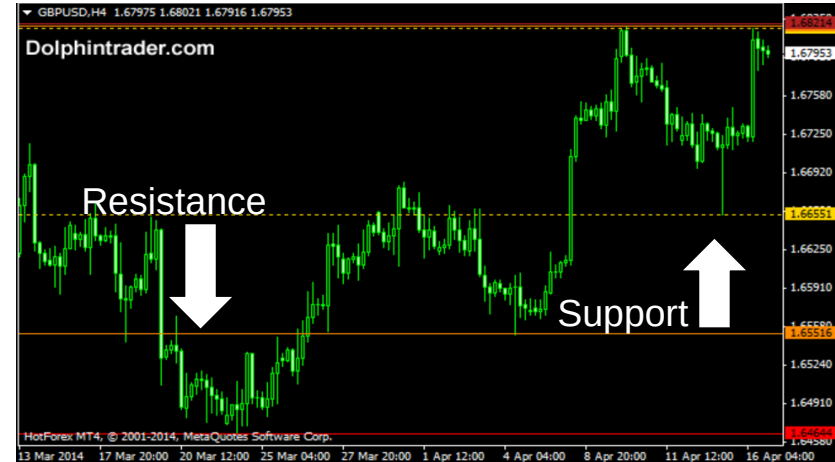
- **Relative Strength Index (RSI)**

Momentum Indicator that shows speed and strength of price

RSI < 30 → Over Sold

RSI > 70 → Over bought

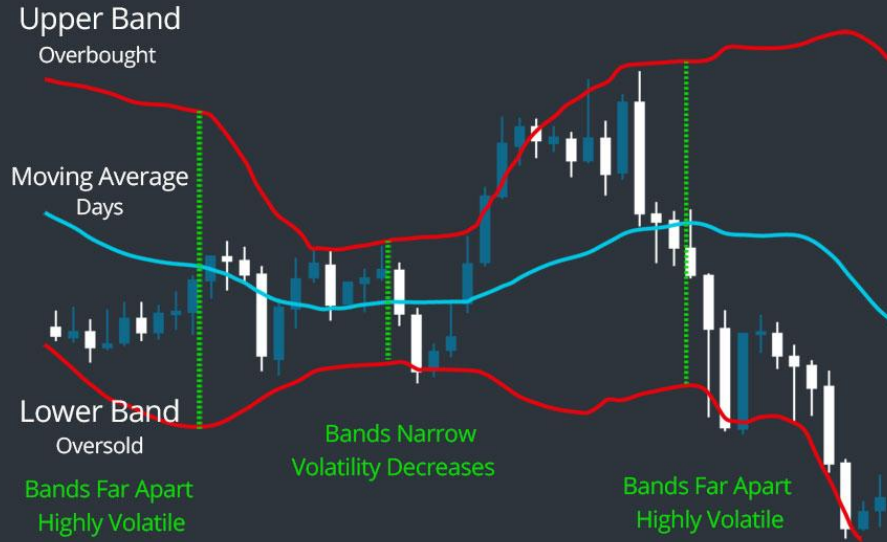
- **Support and Resistance**



Technical Indicators

- [Liquidation chart](#)
- [Market Cap](#)
- Funding Rate

BOLLINGER BANDS AND VOLATILITY



Technical Indicators

- **Bollinger Bands**
 - Upper Band
 - Base Band (20 Periods SMA)
 - Lower Band

Technical Indicators

- **Trend Lines**
 - Key Levels formed by the Market
 - High probability of winning trade

[Trend Lines Link](#)

Trend Patterns

Continuation Pattern

Breakout Pattern

Continuation Pattern



Breakout Pattern



Trend Patterns



Trend Patterns

Common Questions:

Should We draw them hitting the Wick's End, Candle's Close, Candle's Body

Answer:

Draw them where it can give us the most amount of touches

Trend Lines



Trend Lines



Continuation Pattern



Continuation Pattern



Confirmation of Trends with Additional Parameters



Confirmation of Trends with Additional Parameters



Break out Patterns



Break out Patterns



Chart Patterns

- **Double Top**
- **Double Bottom**
- **Head and Shoulder Structure**

Double Top



Double Bottom



Bitcoin head and shoulders pattern forming
(BTCUSD TradingView, 1-day chart)



Chart Patterns

- **Head and Shoulder Structure**

Relative Strength Index



Relative Strength Index



How to automate price prediction using these indicators?

[TradingView](#)

Activity

Pickup a coin analyze the key indicators and chart patterns

Predict the price

Module-7

SENTIMENT AND ON-CHAIN ANALYSIS

Sentiment Analysis

- Twitter Sentiment – Trump, Elon Musk, Blackrock etc
- News Impact
- Fear and Greed Index
- FOMC Meeting
- CPI/PPI Data
- Cryptocurrency Specific Events
 - BTC Halving
 - Eth Upgrades
 - Major Exchange Listings
- Macro Events
 - Geo Political Tension
 - Bank Collapses or Financial Crisis

On chain Analysis

On-chain analysis looks directly at blockchain data (transactions, addresses, etc.) to assess investor behavior.

Steps for On-Chain Analysis:

Check Active Addresses & Transactions:

Increasing active addresses = more network activity (bullish).

Tools: Glassnode, IntoTheBlock, CryptoQuant.

Whale Activity:

Monitor wallets holding 1,000+ BTC. Track whale inflows/outflows on Glassnode or WhaleAlert.

Activity

Perform Sentiment Analysis

Module-8

RISK MANAGEMENT AND TRADING
PSYCHOLOGY

Risk Management in Trading

- **Volatility:** Understanding and managing the inherent risk of volatile markets.
- **Setting stop-loss orders and take-profit levels.**
- **Diversification:** Not putting all your funds in one asset.
- Avoiding emotional trading and FOMO (Fear of Missing Out).
- Using only what you can afford to lose in crypto investments.

Common Mistakes to avoid

- Overtrading
- Chasing losses or overleveraging.
- Falling for “pump-and-dump” schemes.
- Lack of a solid trading plan.
- Failing to understand the technology behind the coin you are trading.

Trading Styles

Scalping

Intraday Trading

Swing Trading

Position Trading

Position Strategies

ETHUSDT Perp ▾ 📊 📉 ...

-1.79%

Open Long Buy 1576.3 523.33K

Max Open 75.70 USD 1576.2 243.07K

Cost 0.00 USD 1576.1 564.97K

Open Short Sell 1576.0 253.70K

0,1 📊

Trade USDC Contracts with 0 Fees
For all users, limited time only!

Open Orders (0) **Positions (1)** Futures Grid 📊

☐ Hide Other Pairs **Close All Positions**

B ETHUSDT Perp Cross 26x 📊

Unrealized PNL (USD) ROI

-2.75 **-64.73%**

Size (USD)	Margin (USD)	Margin Ratio
110.39	4.25	6.13%

Entry Price (USD)	Mark Price (USD)	Liq. Price (USD)
1616.29	1577.31	1480.38

Leverage TP/SL Close Reverse

SOLUSDT Perp ▾ 📊 📉 ...

+1.80%

For all users, limited time only.

Open Orders (2) **Positions (2)** Futures Grid 📊

☐ Hide Other Pairs **Close All Positions**

Unrealized PNL (USD) ROI

-2.58 **-74.71%**

Size (USD)	Margin (USD)	Margin Ratio
90.00	3.46	8.37%

Entry Price (USD)	Mark Price (USD)	Liq. Price (USD)
1624.28	1578.58	1453.15

Leverage TP/SL Close Reverse

S SOLUSDT Perp Cross 25x 📊

Unrealized PNL (USD) ROI

1.06 **+46.01%**

Size (USD)	Margin (USD)	Margin Ratio
58.09	2.32	8.37%

Entry Price (USD)	Mark Price (USD)	Liq. Price (USD)
134.4400	132.0100	148.1122

TP/SL: -- / 136.2000 📊

Leverage TP/SL Close Reverse

SOLUSDT Perp Chart

SOLUSDT Perp ▾ 📊 📉 ...

Price Info Trading Data Square

Last Price ▾ **132.04** 24h High 136.26 24h Vol(SOL) 28.00M

R37,029.30 +1.65% 24h Low 129.40 24h Vol(USD) 3.72B

Mark Price 132.06

Time 15m 1h 4h 1D More ▾ Depth 📊

136.55 136.20 135.28 134.44 134.01 132.73 132.04 131.26 130.48 130.19 8,126,186.17 7,858,697.40 771K 82.1K

PNL +1.05 -58.1 TP/SL Limit 57.6400 131.00

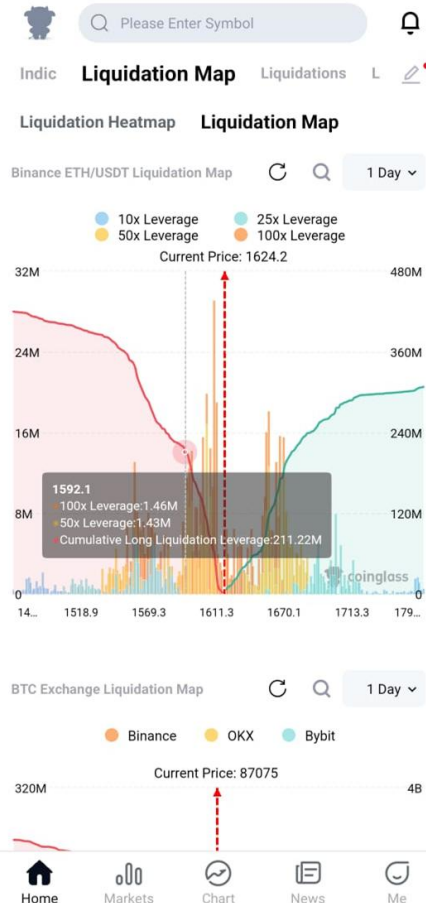
O.I: 8.02M O.I. NV: 1.04B

MA(5): 343,224.5200 MA(10): 344,877.9630

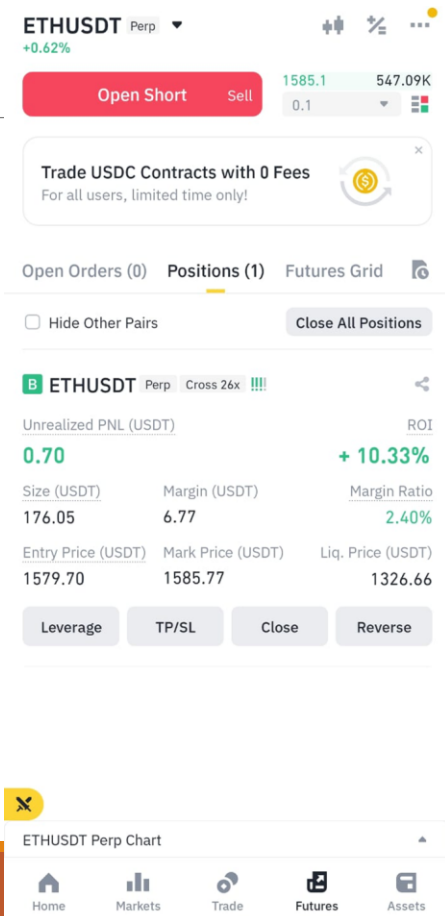
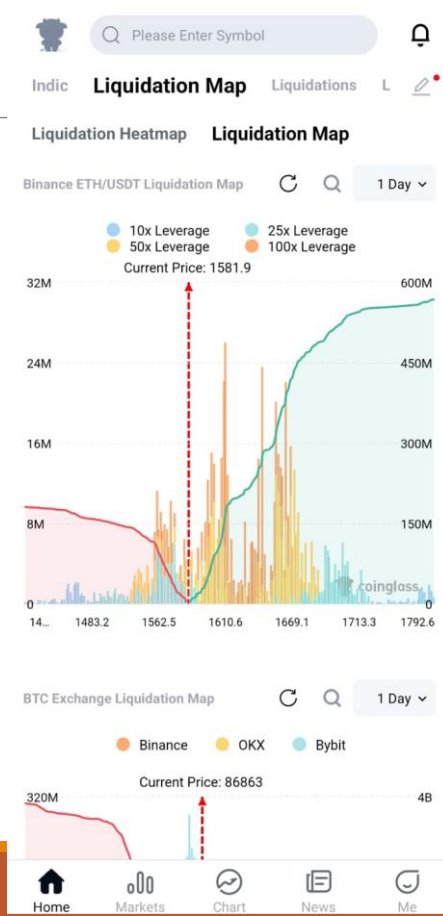
Vol: 304,388.7199

Open Close

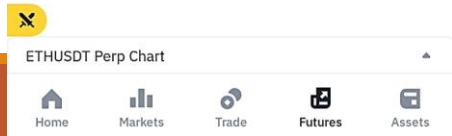
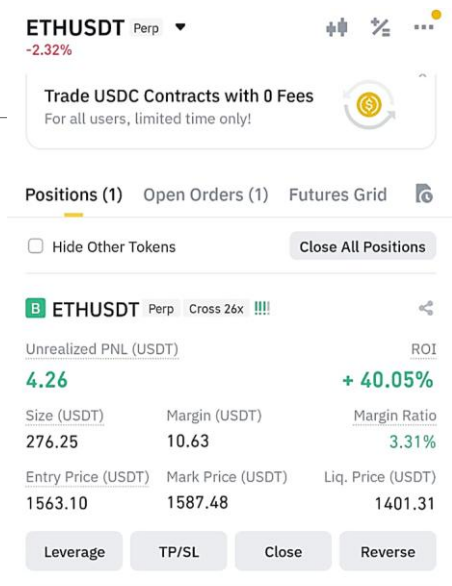
Position Strategies (The Bull/Bear Trap)



Position Strategies (Catch the Swing)



Position Strategies (Catch the Swing)



Some Real Stories

USD🇸🇲	
Asset	
USDT	2025-04-20 16:00:03
Symbol	VOXELUSDT Perpetual
Amount	+151.6502656
USDT	2025-04-20 16:00:02
Symbol	HIGHUSDT Perpetual
Amount	+7.4168574
USDT	2025-04-20 16:00:01
Symbol	NKNUSDT Perpetual
Amount	+31.44836333
USDT	2025-04-20 15:00:00
Symbol	VOXELUSDT Perpetual
Amount	+148.3367808
USDT	2025-04-20 14:00:00
Symbol	VOXELUSDT Perpetual
Amount	+117.87672497
USDT	2025-04-20 12:00:00
Symbol	VOXELUSDT Perpetual
Amount	+139.9830656
USDT	2025-04-20 10:00:00

USD🇸🇲	
Orders (0)	Order History
Position History	Trade History
Transacti	
Mode	
Last updated: 2025-04-18 18:45:46	
* Due to data complexity, there may be some delay. Please scroll down to refresh and update the data.	
-53.01	31737 1000PEPE
Entry Price	Avg. Close Price
0.0092400	0.0075690
Max Open Interest	31737 1000PEPE
Opened	2025-02-23 20:40:56
Closed	2025-02-25 12:29:01
1000BONKUSDT	Perp Cross
Closed	
Closing PNL	Closed Vol.
-166.30	39577 1000BONK
Entry Price	Avg. Close Price
0.0177400	0.0135380
Max Open Interest	39577 1000BONK
Opened	2025-02-10 19:13:22
Closed	2025-02-25 03:03:28
1000FLOKIUSDT	Perp Cross
Closed	
Closing PNL	Closed Vol.
-55.33	7665 1000FLOKI
Entry Price	Avg. Close Price
0.0909000	0.0836800
Max Open Interest	7665 1000FLOKI
Opened	2025-02-24 08:50:01
Closed	2025-02-25 03:03:02
DOGEUSDT	Perp Cross
Closed	
Closing PNL	Closed Vol.
-202.49	5021 DOGE
Entry Price	Avg. Close Price
0.254000	0.213670
Max Open Interest	5021 DOGE

Exchange	Wallet
Overview	Spot
Funding	Earn
Futures	
USD🇸🇲	COIN-M
Margin Balance	
\$0.000000	USD
Today's Realized PnL	-\$6.92(-99.89%)
Wallet Balance (USD)	Unrealized PNL (USD)
\$0.000000	\$0.000000
Trade	Swap
Transfer	
BNB Discount 10%	
BFUSD Earning	APY 2.32%
Positions	Assets
You have no positions.	
Home	Markets
Trade	Futures
Assets	



Price Predictions Platforms

- CryptET <https://cryptet.com/en/>
- AI Models like ChatGPT
- Telegram Channels

Other Methods of Earning

- Trading Bots
- Staking – Passive Income
- Etc.

Activity

Pick a coin and make a list, when to invest and how much to invest

Which position to take and why?

Module-9

DEFI AND ALTCOINS OVERVIEW
INCLUDING MEME COINS

Decentralized Finance (DeFi)

 **1. Decentralized Exchanges (DEXs):**
Trade tokens directly (e.g., Uniswap, SushiSwap)

 **2. Lending & Borrowing:**
Earn interest or borrow crypto (e.g., Aave, Compound)

 **3. Stablecoins:**
Price-stable cryptocurrencies (e.g., DAI, USDC)

 **4. Yield Farming & Liquidity Mining:**
Earn rewards by providing liquidity to DeFi platforms

  **5. Insurance Protocols:**
Smart contract-based insurance (e.g., Nexus Mutual)

 **6. Synthetic Assets:**
Tokenized stocks, commodities, and more (e.g., Synthetix)

Altcoins

Cryptocurrency other than Bitcoin

Categories of Altcoins:

 **Stablecoins:** Pegged to fiat (e.g., USDT, USDC)

 **Utility Tokens:** Used within platforms (e.g., LINK, UNI)

☐ **Smart Contract Platforms:** Compete with Ethereum (e.g., Solana, Cardano, Avalanche)

 **Payment Coins:** Focus on fast, low-cost transfers (e.g., Litecoin, Dash)

 **Game/Metaverse Tokens:** Used in virtual economies (e.g., MANA, AXS)

Meme Coins

- Meme coins are a category of cryptocurrencies that are created based on internet memes, trends, or pop culture references rather than real-world utility or innovation.
- Often launched as a joke or parody, but some have gained massive popularity and value due to social media hype.

Famous Examples: TRUMP, DOGE and SHIB

Key Characteristic:

Community Driven

Platform: pump.fun

Activity

Launch your Meme Coin

Module-10

HANDS-ON TRADING SIMULATIONS

Module-11

SECURITY, SCAMS AND REGULATIONS

Crypto Regulations

1. Legal Status of Cryptocurrencies

- Some countries treat crypto as:
 - **Legal tender** (e.g., El Salvador for Bitcoin)
 - **Property/Asset** (e.g., USA—taxed as property)
 - **Illegal or banned** (e.g., China)

2. Licensing & Registration

- Many jurisdictions require:
 - **Crypto exchanges** and **wallet providers** to register with regulators.
 - Example: **Financial Conduct Authority (UK)**, **SEC (USA)**.

Crypto Regulations

3. Anti-Money Laundering (AML) & Know Your Customer (KYC)

- Exchanges must verify users' identities to prevent:
 - Money laundering
 - Terrorist financing
 - Illegal activities

✓ Example:

- Binance, Coinbase, and others require **ID verification** to comply with **KYC** rules.

Crypto Regulations

4. Taxation

- Crypto gains are often taxable:
 - **Capital gains tax** on profits from trading.
 - **Income tax** for mining or earning crypto.

✓ Example:

- In the **US**, the IRS requires all crypto transactions to be reported.

5. Securities Law

- Some tokens (especially **ICOs** or **security tokens**) are treated as **securities**.
- Subject to **strict regulations** under bodies like the **SEC (U.S.)**.

Crypto Regulations

6. Stablecoins & CBDCs

- Stablecoins (like USDT, USDC) face increased scrutiny for **reserves transparency**.
- Many governments are developing **Central Bank Digital Currencies (CBDCs)** (e.g., Digital Yuan, Digital Euro).

7. DeFi and NFTs

- **DeFi platforms** and **NFT marketplaces** are increasingly under regulatory spotlight due to:
 - Lack of centralized control
 - Potential for fraud, rug pulls, and wash trading

Rug pull

Developers suddenly **withdraw all liquidity** or funds from a project, causing the token's value to crash



Liquidity Rug: Developers remove liquidity from DEX (e.g., Uniswap)

- ❑ Token Minting Abuse: Creators mint more tokens and dump them



Malicious Code: Smart contracts with hidden backdoors :

Red Flags to Watch:

- ✗ No code audit
- 😬 Anonymous dev team
- ▶ Huge token supply held by creators
- 📉 Extremely high APYs (unsustainable)
- ❑ No real use case or whitepaper

Phishing

Fraudulent attempt to steal sensitive information (like private keys or seed phrases)

Common Crypto Phishing Tactics:

- ✉✉ **Fake Emails or Messages:** Pretending to be from wallets, exchanges, or airdrops
- 🌐🌐 **Spoofed Websites:** Look-alike domains to steal login details (e.g., binancee[.]com)
- ➔📱📱 **Fake Wallet Apps:** Uploaded to app stores to trap users
- ☎☎ **Social Engineering:** Impersonating support to extract keys or passwords
- 🔗🔗 **Malicious Links or QR Codes:** Shared on Telegram, Discord, or Twitter

How to Stay Safe:

- ❑ **Never share your seed phrase or private key**
- ✓ **Use official links** (bookmark sites like <https://metamask.io>)
- 🔒🔒❑ **Enable 2FA** and use hardware wallets for added security

Spoofing

Spoofing is a **market manipulation tactic** where traders place large buy/sell orders they don't intend to execute — to **create a false sense of market demand or supply** and mislead other traders.

How Spoofing Works:

- ❑ **Fake Large Orders** are placed (e.g., huge sell wall)

- 📉 Other traders panic and react (price drops)

- ↻ Spoofers cancel fake orders and execute real trades at a better price

- 💰 Profit is made from the manipulated price movement

How to Stay Cautious:

Watch for **unusual order book activity**

Don't base trades on **single large walls**

Use **volume indicators** and **multi-exchange data** to validate trends

Activity

Read recent Rug pulls, Phishing and Spoofing

Module-12

Capstone Project/ Final
Assessment

Create Comprehensive
Trading Strategy

Peer Presentations and
Critiques

Tools and Platforms

- [Binance](#)
- [Bitget](#)
- [OKX](#)
- [CoinGlass](#)
- [Coin Market Cap](#)
- [Coin geko](#)
- Youtube Channels
<https://www.youtube.com/@ilmeaalim>
<https://www.youtube.com/@vnzabbar>
- [Trading View](#)
- [SoSo Value](#)
- [Coin Codex](#)
- [Crypto Quant](#)
- [Strategic Eth Reserve](#)